Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 13

B.Pharma (Sem.-4)

MEDICINAL CHEMISTRY-I

Subject Code : BP-402T

M.Code : 75844

Date of Examination : 02-01-23

Time : 3 Hrs. **Max. Marks : 75**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. Explain the Terms :

a) What are Isosteres? Give example of biological isostere.

b) Give the importance of partition coefficient in drug distribution.

c) Give an example of Phase I metabolic reaction.

d) Give mechanism of action and uses of Tolazoline.

e) Justify the multiple pharmacological actions of acetylcholine.

f) Give mechanism of action and therapeutic uses of organo-phosphorus compounds.

g) Give two examples of phenothiazines acting as antipsychotic agents.

h) Give structure and chemical name of mefanemic acid.

i) What are dissociative anesthetic agents? Give example.

j) Give various therapeutic uses of cholinergic agents.

SECTION-B

2. Classify Anticholinergic agents with examples. Explain detailed chemistry, including stereochemical aspects, of naturally occurring class of anticholinergic agents.

3. Classify antiepileptic agents giving structure of at least one drug from each class. Explain the SAR, mechanism of action and uses of barbiturates.

4. Discuss in detail various physico-chemical and stereochemical aspects affecting the biological activity of drugs.

SECTION-C

5. Discuss various factors affecting drug metabolism.

6. Write a short note on indirect acting sympathomimetic agents.

7. Comment upon various types of cholinergic receptors, their occurrence in body, and their agonists and antagonist.

8. Discuss the synthesis and therapeutic uses of diazepam.

9. Write a short note on ring analogs of phenothiazines.

10. Classify anti-inflammatory agents giving examples, structures and chemical name of atleast two drug from each class.

11. Comment upon cardioselective β-blockers.

12. Discuss SAR of morphine analogs.

13. Give structure, mechanism of action and synthesis of phenylephrine.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.